

Xian-Yu Wang

Last update: October 1, 2026

PhD in Astrophysics

Indiana University, Bloomington, IN 47405

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CURRENT POSITION

Indiana University

Feb. 2023 - Present

Sullivan Prize Postdoctoral Fellow

EDUCATION

University of Chinese Academy of Sciences (UCAS)

Beijing, China

National Astronomical Observatories, Chinese Academy of Sciences (NAOC)

Ph.D in Astrophysics

Jan. 2023

Thesis: *Study of exoplanets based on high-precision photometric and spectroscopic observations*

Advisor: Zhen-Yu Wu

Shandong University

Weihai, Shandong, China

B.S in Space Science and Technology

Jun. 2018

RESEARCH INTERESTS

My research focuses on the architectures and dynamical evolution of exoplanetary systems, leveraging dynamical imprints such as stellar obliquity and orbital eccentricity, together with unified physical modeling and large-scale, data-driven statistical analyses, to uncover their formation and migration histories.

AWARDS

2025	Sullivan Prize Postdoctoral Fellowship
2023	UCAS Outstanding Ph.D. Graduate Award
	Awarded to the top 5% of Ph.D. graduates from the same graduating year at UCAS
2023	Beijing Outstanding Ph.D. Graduate Award
	Awarded to the top 5% of Ph.D. graduates from the same graduating year in Beijing
2023	National Scholarship
	Highest scholarship given by Ministry of Education of the People's Republic of China
2022	Joint PhD Training Program Scholarship, UCAS
	Scholarship provided by UCAS to fund one year-long visit to Indiana University
2021	Pacemaker to Merit Student, NAOC
	Awarded to the top 1% of students
2019	Excellent Student Leader, NAOC
2019	Merit Student, NAOC
	Awarded to the top 20% of students
2018	National Astronomical Observatory Scholarship
	Scholarship for outstanding undergrads in astronomy research
2018	Outstanding Undergraduate Thesis Award, Shandong University
	Awarded to 1 student per major
2015-18	Academic scholarship, Shandong University
	Awarded to the top 30% of students

OBSERVING PROGRAMS

Magellan, PFS, NSF NOIRLab (2026A), 1 night,

PI: Stellar Obliquity of the First Candidate Misaligned Brown Dwarf

Gemini, Zorro, NSF NOIRLab (2026A), 1.55 nights, PI: Binary Friends of Misaligned Hot Jupiters
TNG, HARPS-N, AOT52 (2025B), 10 hours,

Scientific PI: Probing stellar obliquities in the underpopulated Warm Sub-Saturn population

CAHA, CARMENES, 2025B, 1 nights, Scientific PI: Solving the Stellar Obliquity of sub-Saturns

ESO 3.6m, HARPS, P116, 1 nights,

PI: Explore the Unknown: The Stellar Obliquity of Sub-Saturns Around Hot Stars

JWST Cycle 4, 59.6 Primary Spacecraft Hours, CoI:

The Warm Jupiter Opportunity for Understanding Giant Exoplanet Evolution

WIYN, NEID, 2025B, 11 nights, PI: Toward Understanding the 3D Architecture of Warm Giants

WIYN, NEID, 2025A, 12 nights, PI: Characterizing Angles for Warm Giants

WIYN, NEID, 2024B, 12 nights, PI: 3D Architecture of Warm Jupiters

TNG, HARPS-N, 2023B, 1 nights, Scientific PI: HD 80606 b's Spin-Orbit Misalignment Revisited

WIYN, NEID, 2022B, 5 nights, Co-I: Probing Stellar Obliquities with NEID

LCOGT-1m, 2020A, 30 hours, PI, Photometric Follow-Up of Apparent Decaying Orbital WASP-12 b

LCOGT-1m, 2021A, 30 hours, PI, Understanding the shortening period of WASP-12 system

STUDENTS MENTORING

Undergraduate:

Colin G Delaney (second-year undergraduate student at Indiana University) 2026 summer

Colton Romines (second-year undergraduate student at Indiana University) 2024 - 2024

Jace Rusznak (now graduate student at Penn State University) 2023 - 2025

co-supervised with Songhu Wang

Graduate

Emma Dugan (graduated student at Indiana University) 2023 - 2025

co-supervised with Songhu Wang

Jiamei Yang (now PhD Candidate at Beijing Normal University) 2021

PUBLICATIONS

(10 First or Corresponding* Author, 7 Second Author, 41 in total, H index=18, Citations~820, )

First Author:

1. *Warm Sub-Saturns Orbiting Single Stars Are Spin-Orbit Aligned* 
Xian-Yu Wang and Songhu Wang, 2026, **The Astrophysical Journal Letters**, 1008, L32.
2. *Distinct $e - \lambda$ Trends in Three Gas-Giant Mass Regimes* 
Xian-Yu Wang, Songhu Wang, and Konstantin Batygin. 2026, **The Astrophysical Journal Letters**, 1008, L29. Database: www.stellarobliquity.com.
3. *Unified Kraft Break at ~ 6500 K: A Newly Identified Single-Star Obliquity Transition Matches the Classical Rotation Break* 
Xian-Yu Wang, Songhu Wang, and J. M. Joel Ong. 2026, **The Astrophysical Journal Letters**, 996, L7
4. *Early Evidence for Polar Orbits of Sub-Saturns Around Hot Stars* 
Emma Dugan, **Xian-Yu Wang***, Agustin Heron et al. 2025, **The Astrophysical Journal Letters**, 994, L23, as featured by **AAS Press Release**.
5. *From Misaligned Sub-Saturns to Aligned Brown Dwarfs: The Highest M_p/M_* Systems Exhibit Low Obliquities, Even around Hot Stars* 
Jace Rusznak, **Xian-Yu Wang***, Malena Rice, Songhu Wang, **The Astrophysical Journal Letters**, 983, L42, as featured by **AAS Nova**.

6. *Single-Star Warm-Jupiter Systems Tend to Be Aligned, Even Around Hot Stellar Hosts: No T_{eff} - λ Dependency* 
Xian-Yu Wang, Malena Rice, Songhu Wang, et al. 2024, **The Astrophysical Journal Letters**, 973, L21, as featured by **AAS Nova**.
7. *The Aligned Orbit of WASP-148b, the Only Known Hot Jupiter with a Nearby Warm Jupiter Companion, from NEID and HIRES* 
Xian-Yu Wang, Malena Rice, Songhu Wang, et al. 2022, **The Astrophysical Journal Letters**, 926, L8
8. *Transiting Exoplanet Monitoring Project (TEMP). VI. The Homogeneous Refinement of System Parameters for 39 Transiting Hot Jupiters with 127 New Light Curves* 
Xian-Yu Wang, Yong-Hao Wang, Songhu Wang, et al. 2021, **The Astrophysical Journal Supplement Series**, 255, 15
9. *Transiting Exoplanet Monitoring Project (TEMP). IV. Refined System Parameters, Transit Timing Variations and Orbital Stability of the Transiting Planetary System HAT-P-25* 
Xian-Yu Wang, Songhu Wang, Tobias Hinse, et al. 2018, **The Publications of the Astronomical Society of the Pacific**, 130, 064401
10. *New analysis of the fraction of observable nights at astronomical sites based on FengYun-2 satellite data* 
Xian-Yu Wang, Zhen-Yu Wu, Jing Liu, et al. 2022, **Monthly Notices of the Royal Astronomical Society**, 511, 4

Second Author:

11. *The Discovery of K2-232c: Divergent Formation Histories for Hot and Warm Jupiters Based on Outer Companion Eccentricity* 
 Jessica Ranshaw, **Xian-Yu Wang**, Jennifer Burt et al. 2026, accepted for publication in **The Astronomical Journal**, 532, 604
12. *The SNR of idealised radial velocity signals* 
 David Kipping & **Xian-Yu Wang**. 2024, **Monthly Notices of the Royal Astronomical Society**, 532, 604
13. *TOI-1670 c, a 40 day Orbital Period Warm Jupiter in a Compact System, Is Well Aligned* 
 Jack Lubin, **Xian-Yu Wang**, Malena Rice, et al. 2023, **The Astrophysical Journal Letters**, 295, 5
14. *Evidence for Low-Level Dynamical Excitation in Near-Resonant Exoplanet Systems* 
 Malena Rice, **Xian-Yu Wang**, Songhu Wang, et al. 2023, **The Astronomical Journal**, 166, 266
15. *The Spin-Orbit Misalignment of TOI-1842b: The First Measurement of the Rossiter-McLaughlin Effect for a Warm Sub-Saturn around a Massive Star* 
 Kyle Hixenbaugh, **Xian-Yu Wang**, Malena Rice, Songhu Wang, 2023, **The Astrophysical Journal Letters**, 949, 35, as featured by **AAS Nova**.
16. *Photometric follow-up observations and transit timing analysis of HAT-P-37b* 
 Jia-Mei Yang, **Xian-Yu Wang**, Kai Li, et al. 2021, **Publications of the Astronomical Society of Japan**, 73, 1010
17. *Transiting Exoplanet Monitoring Project (TEMP). I. Refined System Parameters and Transit variations of HAT-P-29* 
 Songhu Wang, **Xian-Yu Wang**, Yong-Hao Wang, et al. 2018, **The Astronomical Journal**, 156, 181

Contributing Author:

18. *Stellar Companions Sculpt Hot Jupiter Formation and Spin-Orbit Evolution* 
 Cheyanne Shariat, Kareem El-Badry, Songhu Wang, **Xian-Yu Wang**, et al. 2026, **arXiv e-prints**, arXiv:2609.19249.

19. *Predicting Macroturbulence in F/G/K Dwarfs to 100 m/s Precision* 
Jack Lubin, Erik Petigura, Kento Masuda and 6 coauthors including **Xian-Yu Wang**, 2026, **arXiv e-prints**, arXiv:2609.17693.
20. *High-Eccentricity Tidal Migration Driven by Secular Chaos in Wide-Binary Systems* 
Yurou Liu, Hareesh Gautham Bhaskar, **Xian-Yu Wang**, Cristobal Petrovich, 2026, **arXiv e-prints**, arXiv:2606.17344.
21. *The OATMEAL Survey. II. The 3D spin-orbit obliquity of an eccentric transiting brown dwarf in the Ruprecht 147 open cluster* 
Theron W. Carmichael, Steven Giacalone, Noah Vowell and 11 coauthors including **Xian-Yu Wang**, 2026, **The Astronomical Journal**, 171, 12.
22. *Spin-Orbit Alignment of Two Neptune-size Planets Younger than 500 Myr: TOI-560 b and TOI-5082 b* 
Elina Y. Zhang, Fei Dai, Andrew W. Howard and 25 coauthors including **Xian-Yu Wang**, 2026, accepted for publication in **The Astronomical Journal**.
23. *TOI-880 is an Aligned, Coplanar, Multiplanet System* 
Elina Y. Zhang, Huan-Yu Teng, Fei Dai and 22 coauthors including **Xian-Yu Wang**, 2025, **The Astronomical Journal**, 170, 3, 175.
24. *Migration and Evolution of giant ExoPlanets (MEEP). II. Super-Jupiters and Lithium-rich host stars* 
Jack Schulte, Joseph E Rodriguez, David W. Latham and 29 coauthors including **Xian-Yu Wang**, 2025, **Monthly Notices of the Royal Astronomical Society**, 543, 1, 292.
25. *Evidence for Primordial Alignment II: Insights from Stellar Obliquity Measurements For Hot Jupiters in Compact Multi-planet Systems* 
Brandon T. Radzom, Jiayin Dong, Malena Rice, **Xian-Yu Wang**, et al 2025, **The Astronomical Journal**, 169, 189
26. *SOLES XII. The Aligned Orbit of TOI-2533 b, a Transiting Brown Dwarf Orbiting an F8-type Star* 
Thiago Ferreira, Malena Rice, **Xian-Yu Wang**, Songhu Wang, 2024, **The Astronomical Journal**, 168, 145
27. *Evidence for Primordial Alignment: Insights from Stellar Obliquity Measurements for Compact Sub-Saturn Systems* 
Brandon T. Radzom, Jiayin Dong, Malena Rice, **Xian-Yu Wang**, et al 2024, **The Astronomical Journal**, 168, 116
28. *A hot-Jupiter progenitor on a super-eccentric retrograde orbit* 
Arvind F. Gupta, Sarah C. Millholland, Haedam Im and 53 coauthors including **Xian-Yu Wang**, 2023, **Nature**, 632, 8023, 50.
29. *The PFS view of TOI-677 b: A spin-orbit aligned warm Jupiter in a dynamically hot system* 
Qingru Hu, Malena Rice, **Xian-Yu Wang**, et al 2024, **The Astronomical Journal**, 167, 175
30. *The GAPS Programme at TNG L – TOI-4515 b: An eccentric warm Jupiter orbiting a 1.2 Gyr-old G-star* 
I. Carleo, L. Malavolta, S. Desidera, and 68 coauthors including **Xian-Yu Wang**, 2023, **Astronomy & Astrophysics**, 682, 138
31. *SOLES VII: The Spin-Orbit Alignment of WASP-106 b, a Warm Jupiter Along the Kraft Break* 
Josette Wright, Malena Rice, **Xian-Yu Wang**, et al 2023, **The Astronomical Journal**, 166, 217
32. *The Orbital Architecture of Qatar-6: A Fully Aligned Three-body System?* 
Malena Rice, Songhu Wang, Konstantin Gerbig, and 5 coauthors including **Xian-Yu Wang**, 2023, **The Astronomical Journal**, 165, 65
33. *TOI-1136 is a Young, Coplanar, Aligned Planetary System in a Pristine Resonant Chain* 

Fei Dai, Kento Masuda, Corey Beard and 60 coauthors including **Xian-Yu Wang**, 2023, **The Astronomical Journal**, 165, 33

34. *A Tendency Toward Alignment in Single-star Warm-Jupiter Systems* 
Malena Rice, Songhu Wang, **Xian-Yu Wang** et al. 2022, **The Astronomical Journal**, 164, 104
35. *Revisiting the Full Sets of Orbital Parameters for the XO-3 System: No evidence for Temporal Variation of the Spin-Orbit Angle* 
Keduse Worku, Songhu Wang, Jennifer Burt, and 14 coauthors including **Xian-Yu Wang**, 2022, **The Astronomical Journal**, 163, 158
36. *SOLES I: The Spin–Orbit Alignment of K2-140 b* 
Malena Rice, Songhu Wang, Andrew W. Howard, and 8 coauthors including **Xian-Yu Wang**, 2021, **The Astronomical Journal**, 162, 182
37. *The Aligned Orbit of the Eccentric Warm Jupiter K2-232b* 
Songhu Wang, Joshua N. Winn, Brett C. Addison, and 8 coauthors including **Xian-Yu Wang**, 2021, **The Astronomical Journal**, 162, 50
38. *The Youngest Planet to Have a Spin-Orbit Alignment Measurement AU Mic b* 
Brett C. Addison, Jonathan Horner, Brett C. Addison, and 8 coauthors including **Xian-Yu Wang**, 2021, **The Astronomical Journal**, 162, 137
39. *TOI 564 b and TOI 905 b: Grazing and Fully Transiting Hot Jupiters Discovered by TESS* 
Allen B. Davis, Songhu Wang, Matias Jones, and 54 coauthors including **Xian-Yu Wang**, 2020, **The Astronomical Journal**, 160, 229
40. *HD 202772A b: A Transiting Hot Jupiter Around a Bright, Mildly Evolved Star in Discovered by TESS* 
Songhu Wang, Matias Jones, Avi Shporer, and 57 coauthors including **Xian-Yu Wang**, 2018, **The Astronomical Journal**, 157, 51
41. *A possible giant planet orbiting the cataclysmic variable LX Ser* 
Li Kai, Hu Shaoming, Zhou Jilin and 6 coauthors including **Xian-Yu Wang**, 2017, **Publications of the Astronomical Society of Japan**, 69, 28

SELECTED TALKS AND CONFERENCES

Conference Talks:

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|---|-----------|
| * Emerging Researchers in Exoplanet Science Symposium XI, University of Michigan
Distinct Eccentricity - Stellar Obliquity Trends in Three Gas-Giant Mass Regimes | Jul. 2026 |
| * Exoplanet VI, Porto, Portugal
Distinct Eccentricity - Stellar Obliquity Trends in Three Gas-Giant Mass Regimes | Jun. 2026 |
| * Annual Meeting of the Hubei Astronomical Society
Probing Hot Jupiters' Migration Theories Using Photometric and Spectroscopic Observations | Dec. 2025 |
| * International Conference on Exoplanets and Planet Formation (EPF) 2025
Distinct Eccentricity - Stellar Obliquity Trends in Three Gas-Giant Mass Regimes | Dec. 2025 |
| * Great Lakes Exoplanet Area Meeting (GLEAM) 2025
A Homogeneous Stellar Obliquity Sample from Global Modeling of Rossiter–McLaughlin Measurements | Nov. 2025 |
| * Know Thy Star, Know Thy Planet 2, Caltech, Pasadena
Single-Star Warm-Jupiter Systems Tend to Be Aligned, Even Around Hot Stellar Hosts:
No $T_{\text{eff}} - \lambda$ Dependency | Feb. 2025 |
| * 245th AAS Annual Winter Meeting, National Harbor, Maryland
Single-Star Warm-Jupiter Systems Tend to Be Aligned, Even Around Hot Stellar Hosts:
No $T_{\text{eff}} - \lambda$ Dependency | Jan. 2025 |

* TESS Science Conference III (plenary session), MIT Warm Jupiters Tend to Be Aligned, Even Around Hot Stars,	Aug. 2024
* Emerging Researchers in Exoplanet Science Symposium IX, Cornell University Warm Jupiters Tend to Be Aligned, Even Around Hot Stars,	Jul. 2024
* Exoplanet V, Netherland Tri-Angles: 0° , 90° , 180° — Hot Jupiters' Signature Spin-Orbit Angles	Jun. 2024
* 243th AAS Annual Winter Meeting (thesis talk), New Orleans Violence or Quiescence Probing Hot Jupiters' Migration Theories Using Photometric and Spectroscopic Observations	Jan. 2024
* Great Lakes Exoplanet Area Meeting (GLEAM) 2023 Three population in stellar obliquity distribution of hot Jupiters: 0, 90, and 180 degrees	Oct. 2023
* Asia Oceania Geosciences Society (AOGS) Annual Meeting, Observational and Theoretical Aspects of Exoplanets, Singapore <i>3D Configuration of a Compact Multi-giant System Lying at the Stability Boundary</i>	Aug. 2023
* Emerging Researchers in Exoplanet Science VII, The Pennsylvania State University <i>The Aligned Orbit of WASP-148b and the statistic implications from the distribution of stellar sky-projected obliquities</i>	Aug. 2022
* Annual Conference of the Chinese Astronomical Society, Nanchong, China <i>Transiting Exoplanet Monitoring Project (TEMP)</i>	Dec. 2021
* Annual Conference of the Chinese Astronomical Society, Nanchong, China <i>The Aligned Orbit of WASP-148b</i>	Dec. 2021

Seminar and Lunch Talks:

* Astronomy Seminar, Westlake University, Zhejiang	June. 2025
* ZJU IA Spring Seminar, Zhejiang University, Zhejiang	June. 2025
* TDLI Seminar, Tsung-Dao Lee Institute, Shanghai	June. 2025
* SHAO lunch talk, Shanghai Astronomical Observatory, Shanghai	June. 2025
* PMO Youth Forum, Purple Mountain Observatory, Nanjing	June. 2025
* Exoplanet lunch talk, Nanjing University, Nanjing	June. 2025
* Astrop-ph Seminar, Peking University, Beijing	May. 2025
* PDS Seminar, Tsinghua University, Beijing	May. 2025
* Youth Innovation Promotion Association academic Seminar, NAOC, Beijing	May. 2025
* CIERA - Northwestern University, Evanston, IL	Mar. 2025
* Harvard CfA Exoplanet Pizza Lunch, Cambridge, MA	Nov. 2024
* David Charbonneau Group Meeting, Cambridge, MA	Nov. 2024
* Tea Talk, Indiana University Bloomington	Apr. 2024
* Tea Talk, Indiana University Bloomington	Sep. 2022
* Lunch talk, South-Western Institute For Astronomy Research, Kunming, China	Mar. 2022

Conference Posters:

* Poster, Emerging Researchers in Exoplanet Science Symposium VIII, New Heaven Homogeneous Studies on the Stellar Obliquities	Jun. 2023
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* Poster, 54th Division on Dynamical Astronomy, Michigan
3D configuration of a compact multi-giant system lying at the stability boundary

May. 2023

SERVICE AND OUTREACH

Panelist for NASA Astrophysics Proposal Review

Referee for AJ, ApJ, and ApJL

Referee for ESO Period Call proposals

Referee for Gemini Proposal

Special Issue Editor for “Formation and Evolution of Exoplanets” in *Universe*.

AAS 245, Session Chair for *Extrasolar Planets: Populations II*

Jan. 2025

AAS 243, Session Chair for *Exoplanet Searches and Surveys II*

Jan. 2024

AAS 243, Judge for *Chambliss Astronomy Achievement Student Award* poster competition

Jan. 2024

GLEAM 2023 Conference, Member of SOC & LOC, and Session Chair

Jul. 2023

TEACHING

Indiana University Bloomington

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| * AST-A 100 The Solar System | 2024 |
| * AST-A 100 The Solar System | 2023 |

University of Chinese Academy of Sciences

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| * Stellar Structure and Evolution, Teaching assistance | 2019 |
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Shandong University

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| * Optics, Teaching assistance | 2015 |
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